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9702/32

May/June 2008

2 hours

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

DO NOT WRITE IN ANY BARCODES.

You are reminded of the need for good English and clear presentation in your answers.

All questions in this paper carry equal marks.

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2	
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This document consists of **8** printed pages and **4** blank pages.

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- 1** In this experiment you will measure the period of oscillation of a wooden rod weighted at different positions. You will use the results to determine the mass of the rod.

- (a)** Set up the rod so that it is pivoted by a nail as shown in Fig. 1.1. Another nail should be placed through the bottom hole and a 0.100 kg mass is to be held in place with a small piece of Blu-Tack.

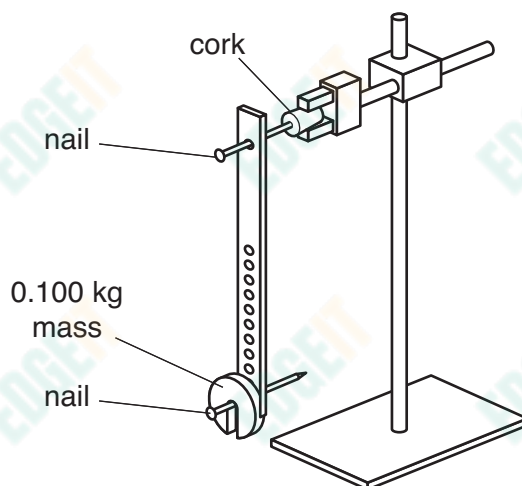


Fig. 1.1

- (b)** Measure the distance l between the two nails.

$l = \dots\dots\dots$ m

- (c)** Displace the rod so that it oscillates with a small amplitude in a vertical plane as shown in Fig. 1.2.

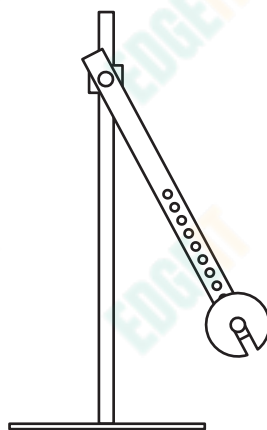


Fig. 1.2

Determine the period T of oscillation.

$T = \dots\dots\dots$ s

- (d) Move the bottom nail and the mass to different holes and repeat (b) and (c) until you have six sets of readings for l and T . Include values for T^2 in your table of results.

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- (e) Plot a graph of T^2 on the y-axis against l on the x-axis and draw the line of best fit.
- (f) Determine the gradient and y-intercept of this line.

gradient =

y-intercept =

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- (g) It is suggested that the relationship between T^2 and l is

$$T^2 = \frac{4\pi^2 ml}{g(m+M)} + z$$

where m is 0.100 kg, M is the mass of the rod, $g = 9.81 \text{ N kg}^{-1}$ and z is a constant for this experiment.

Use your answers from (f) to determine the values of M and z . Give appropriate units.

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$M = \dots\dots\dots$

$z = \dots\dots\dots$

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Please turn over for Question 2

You may not need to use all of the materials provided.

- 2 (a) Roll out the modelling clay into a cylinder of length 4 cm. Wrap the 4 cm width of paper around the cylinder's length and fix with Sellotape as shown in Fig. 2.1.

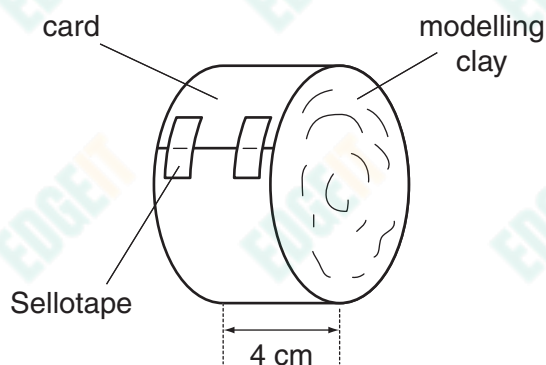


Fig. 2.1

- (b) (i) Use the string to measure the circumference c of the cylinder.

$c =$

- (ii) Estimate the percentage uncertainty in c .

% uncertainty in $c =$

- (iii) Use your answer to (i) to determine the radius r of the cylinder.

$r =$

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- (c) Place the cylinder at the top of the slope and release it so that it rolls down the slope. Measure the time t for the cylinder to travel a distance of 50 cm from rest.

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$t = \dots\dots\dots$ s

- (d) Undo the paper and roll the modelling clay into a cylinder of length 8 cm. Wrap the 8 cm width of paper around the cylinder's length and fix with Sellotape. Repeat (b)(i), (b)(iii) and (c).

$c = \dots\dots\dots$

$r = \dots\dots\dots$

$t = \dots\dots\dots$

- (e) Explain whether your results support the idea that t^2 is proportional to r .

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- (f) (i) State four sources of error or limitations of the procedure for this experiment.

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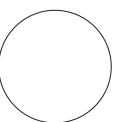
- (ii) Suggest four improvements that could be made to the experiment. You may suggest the use of other apparatus or different procedures.

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